

TrES-5b

R-1500

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130522 · created 2026-08-01T14:43:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456434.874 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.05
Transit depth (Rp/Rs)^2	3.22 +/- 1.79 [%]
Orbital Inclination (inc)	82.14 +/- 2.01
Airmass coefficient 1 (a1)	1.124 +/- 0.04
Airmass coefficient 2 (a2)	-0.087 +/- 0.033
Scatter in the residuals of the lightcurve fit is	3.59 %
Best Comparison Star	#1 - [276, 79]
Optimal Method	PSF photometry
Transit Duration (day)	0.058 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.179 ± 0.05 vs 0.142 (deviation 0.74σ)
Duration fit vs published	0.058 d vs 0.0758 d (deviation 0.81σ)
Mid-time O–C	-3.3 min
Depth SNR	3.6
Residual scatter	3.59 %

Light curve

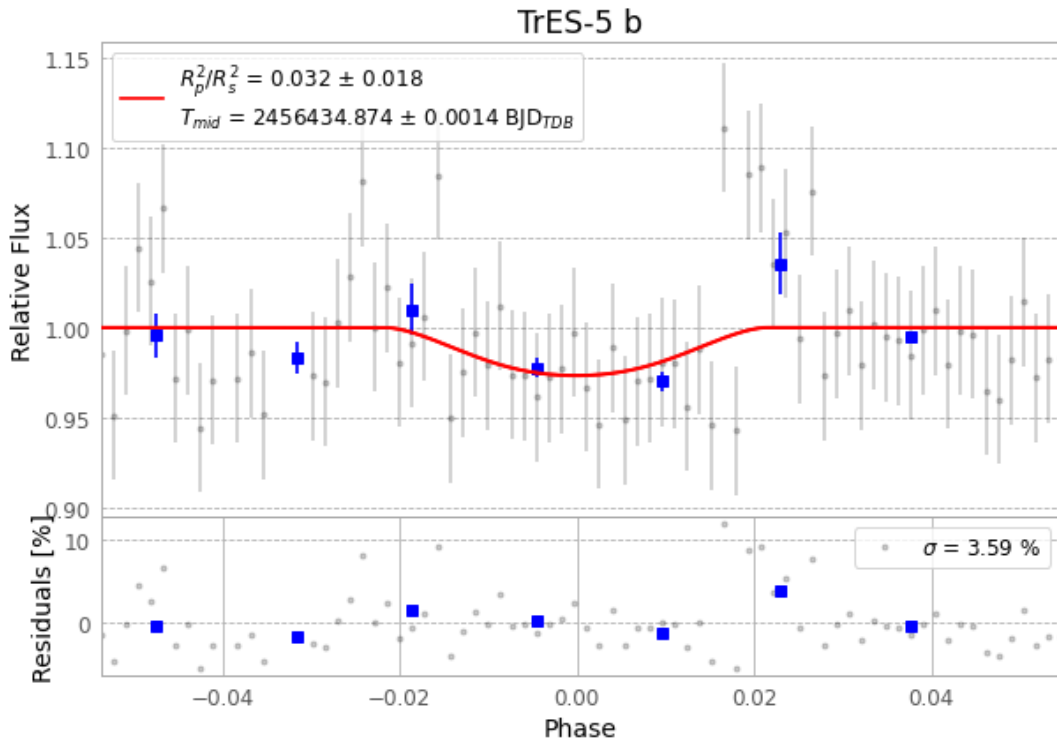


Figure 1 — FinalLightCurve_TrES-5 b_2013-05-22.png (SHA-256 195ff156fa1d1c07...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 176], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d38e6395f80a7f7d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), TRES-5130522070312.FITS → TRES-5130522105413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130522.log (e388dd7053c0...), FinalLightCurve_TrES-5 b_2013-05-22.png (195ff156fa1d...), FinalLightCurve_TrES-5 b_2013-05-22.csv (6a1facbc1c43...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:43Z.